

# Varun Hirethotlukere Shamaraju

Data Engineer | Cloud Data Platforms

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**Work authorisation:** German student permit (§16b), Werkstudent eligible. Post-study permit (§20(3), unrestricted full-time work) from September 2026. **No sponsorship required.**



## Profile

Data engineer with 2 years 7 months of production experience across telecommunications and consulting, now completing an M.Sc. in Computer Science (Big Data and Artificial Intelligence) at SRH University Leipzig. Built and owned ingestion pipelines processing over five million records daily for an ISP serving two million subscribers, and delivered an AWS re-platforming that cut a consulting client's warehouse costs by 15%. Works the full path from source system to production model across Airflow, Terraform, CI/CD and MLOps. Based in Munich with unrestricted work rights from September 2026.

## Core Skills

|                        |                                                                                                                              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Languages & Query      | Python, SQL, Bash                                                                                                            |
| Data Engineering       | Apache Airflow, ETL/ELT pipeline design, Snowflake, PySpark, Kafka, data quality and validation, PostgreSQL, MongoDB         |
| ML & MLOps             | PyTorch, scikit-learn, AWS SageMaker, MLflow, predictive modelling, model versioning, A/B testing, monitoring and retraining |
| Cloud & Infrastructure | AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD pipelines, Git                                                |
| Observability          | Prometheus, Grafana, pipeline alerting and health checks                                                                     |
| Analytics & Delivery   | Power BI, dashboard design, stakeholder reporting, FastAPI, REST                                                             |

## Work Experience

**Analyst, Deloitte India (Deloitte South Asia LLP), Bengaluru** Oct 2024 – Mar 2025

Six-person analytics pod delivering cloud migration and member analytics for a client with a 150,000-member base. Client-facing from week one. Left to commence the M.Sc. programme in Germany.

- Re-platformed 12 legacy data processes onto AWS, improving query performance 25% and reducing data warehouse costs **15%**, retiring the client's overnight batch dependency
- Developed member engagement scoring models in Python and Amazon SageMaker, reaching **0.78 AUC** across 150K members and lifting campaign conversion **9%**, which redirected marketing spend toward the highest-propensity segments
- Built five executive dashboards in Power BI serving 30+ stakeholders across three business units, replacing 20 hours of manual reporting each month
- Presented migration trade-offs and model results directly to client stakeholders including two directors, translating technical findings into commercial recommendations

**Software Engineer, Atria Convergence Technologies (ACT Fibernet), Bengaluru** Sep 2022 – Oct 2024

Team of six data engineers within a 40-person technology group, supporting an ISP with 2M+ broadband subscribers across 20+ Indian cities. Owned ingestion and model deployment for network operations analytics.

- Designed and automated ingestion across three CRM and OSS/BSS systems using OCI Data Integration and Apache Airflow, processing **5M+ records daily** and cutting end-to-end latency from **six hours to under one hour**, giving network operations same-day fault visibility in place of next-day batch reports
- Owned the orchestration layer for 40+ scheduled pipelines, introducing retry logic, alerting and data quality checks that took failed overnight runs from several a week to roughly one a month
- Implemented MLOps practices including model versioning and A/B testing for six production models on OCI, shortening deployment cycles from **three weeks to one week**
- Deployed predictive maintenance models across **1,200+ network nodes**, forecasting failures seven days ahead and reducing unplanned downtime **12%**, which cut emergency field dispatches
- Containerised deployments with Docker and provisioned infrastructure with Terraform, making releases

repeatable across environments; mentored two junior engineers on pipeline design and review standards

## Selected Projects

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- **SIGMA: Equity Research Service**

Self-hosted analysis service pulling market data, fundamentals and NSE/BSE corporate action calendars through a provider interface into SQLite, with cache TTLs, retry and backoff against rate-limited upstreams, exact decimal money arithmetic and configuration that fails at startup rather than at request time. Ships as a single container with GitHub Actions CI and a network-isolated test suite replaying recorded fixtures. *Python 3.12, Docker, SQLite, GitHub Actions, Pytest, MyPy, Ruff.*

- **Serverless AI Assistant (Terraform + AWS Lambda)**

Gemini-powered conversational bot on AWS Lambda, fully provisioned as infrastructure-as-code with Terraform and tested locally against LocalStack before deployment. *AWS Lambda, Terraform, LocalStack, CI/CD.*

- **NSE Index Dashboard**

Interactive Streamlit dashboard tracking NSE indices and equities, backed by a live data feed from Yahoo Finance with caching to keep response times low under repeated queries. *Python, Streamlit, live data ingestion.*

- **SAGE: Stock Analysis and Guidance Engine**

Fundamental stock analysis tool that fetches real-time financial data and generates investment insights using locally deployed LLMs, keeping inference on-premise and avoiding per-token API cost. *Python, local LLM inference, financial data APIs.*

- **Early Detection of Depression from Social Media Data**

Benchmarked multiple classifiers on social media text and deployed the best-performing model behind a web application that scores depression severity for clinical evaluation. Published research, 2025. *Python, NLP, scikit-learn, model benchmarking.*

## Education

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### M.Sc. Computer Science: Big Data and Artificial Intelligence (accelerated track)

2025 – 2026

SRH University, Leipzig, Germany | Grade: **1.6** (German scale, 1.0 = best)

Relevant modules: Machine Learning, Big Data Engineering, Cloud Computing and Distributed Systems, Deep Learning, Data Management.

### B.E. Computer Science and Engineering

2018 – 2022

Visvesvaraya Technological University, Bengaluru, India.

## Certifications and Languages

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- AWS Certified Cloud Practitioner, Amazon Web Services, 2026
- OCI Certified Generative AI Professional, Oracle, 2025
- OCI Certified Data Science Professional, Oracle, 2025

**Languages:** English (C1, professional proficiency) | German (A2) | Kannada (native)